

Week-4

Task 1: Write a C program where a parent process creates multiple child processes and synchronizes their completion using `wait()` and `waitpid()`. Compare the behavior of both functions.

```
root@LAPTOP-RJFD8EBT:~/task5# nano week4_task1.c
root@LAPTOP-RJFD8EBT:~/task5# gcc week4_task1.c -o week4_task1
root@LAPTOP-RJFD8EBT:~/task5# ls -l
total 56
-rwxr-xr-x 1 root root 20704 Aug 17 05:07 task5
-rw-r--r-- 1 root root 4360 Aug 17 05:06 task5.c
-rwxr-xr-x 1 root root 16400 Aug 17 05:16 week4_task1
-rw-r--r-- 1 root root 1933 Aug 17 05:16 week4_task1.c
root@LAPTOP-RJFD8EBT:~/task5# ./week4_task1
Parent Process Started
Parent PID: 712

Child 1 started. PID = 713, Parent PID = 712
Child 2 started. PID = 714, Parent PID = 712

All 3 child processes have been created.

Using wait():
Child 3 started. PID = 715, Parent PID = 712
Child 1 completed. PID = 713
wait() detected child PID 713 finished with exit status 10

Using waitpid():
waitpid failed: No child processes
Child 2 completed. PID = 714
waitpid() detected child PID 714 finished with exit status 11
Child 3 completed. PID = 715
waitpid() detected child PID 715 finished with exit status 12

Parent process completed.
root@LAPTOP-RJFD8EBT:~/task5# |
```

Task 2: Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

```
root@LAPTOP-RJFD8EBT:~/task5# cd ~/OS_Week4
nano zombie.c
-bash: cd: /root/OS_Week4: No such file or directory
root@LAPTOP-RJFD8EBT:~/task5# gcc zombie.c -o zombie
root@LAPTOP-RJFD8EBT:~/task5# ./zombie
Parent process: PID = 778
Child process: PID = 779
Child process: PID = 779
Child PID = 779
Parent sleeping...
Child process terminating...
Parent terminating...
root@LAPTOP-RJFD8EBT:~/task5# |
```

```
Child process terminating...
Parent terminating...
root@LAPTOP-RJFD8EBT:~/task5# ps -el | grep zombie_fixed
root@LAPTOP-RJFD8EBT:~/task5# ps -o pid,ppid,state,cmd -C zombie_fixed
  PID    PPID S  CMD
root@LAPTOP-RJFD8EBT:~/task5# |
```